

Project Design Phase-II

Technology Stack (Architecture & Stack)

Date	30 September 2026
Team ID	SWTID-2026-1542
Project Name	AnswerBot – AI-powered FAQ and Customer Support Automation
Maximum Marks	4 Marks

Technical Architecture:

The technical architecture of AnswerBot is shown below, followed by Table-1 (Components & Technologies) and Table-2 (Application Characteristics).

AnswerBot – Technical Architecture (MVC, headless REST API)

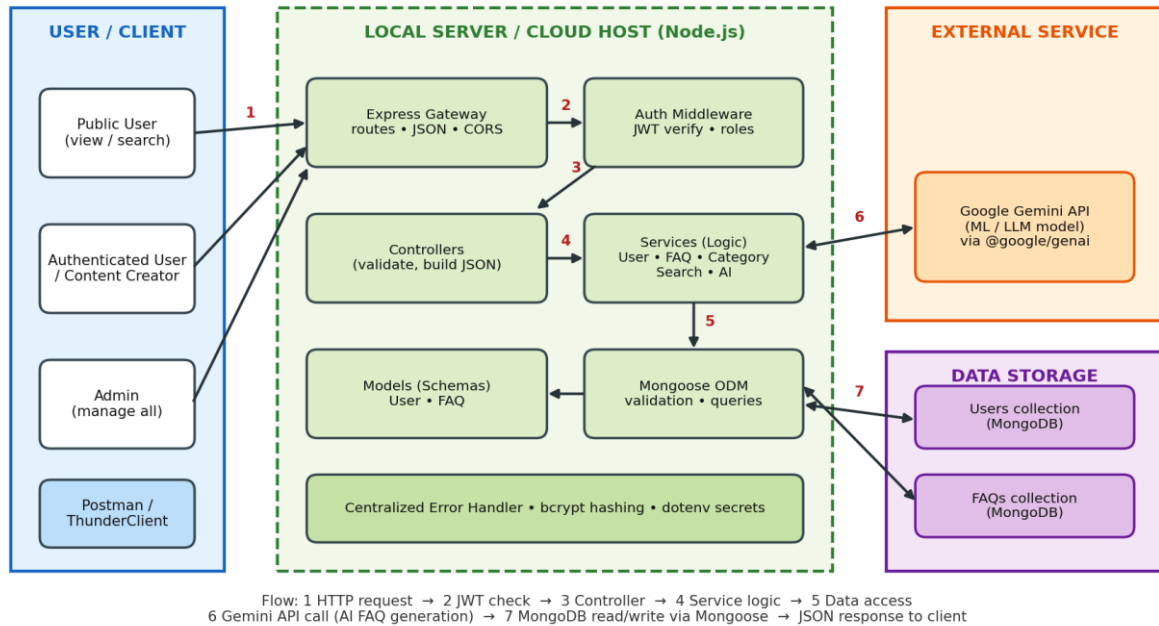


Figure 1: AnswerBot Technical Architecture

Table-1 : Components & Technologies:

S.No	Component	Description	Technology
1	User Interface	How the user interacts with the application. AnswerBot is a headless REST API, so users send HTTP requests and receive JSON responses through an API client.	Postman / ThunderClient (JSON over HTTP)

S.No	Component	Description	Technology
2	Application Logic-1	Express server gateway: listens on the port, maps GET / POST / PUT / DELETE routes, parses JSON payloads and handles CORS.	Node.js, Express.js
3	Application Logic-2	Authentication and authorization: user registration, login, password hashing, JWT verification and role-based access (Admin, Content Creator, User, Public).	jsonwebtoken (JWT), bcrypt
4	Application Logic-3	AI FAQ generation: creates question, answer and category from a topic and supports AI-assisted answers.	Google Gemini SDK (@google/genai)
5	Database	Stores Users and FAQs (question, answer, category, createdBy) with schema validation and defaults.	MongoDB with Mongoose ODM
6	Cloud Database	Managed database service when the application is deployed to the cloud (local MongoDB is used during development).	MongoDB Atlas (optional)
7	File Storage	No user file uploads. Configuration and secrets (DB URI, JWT secret, Gemini API key) are kept in an environment file on the server.	Local filesystem, dotenv (.env)
8	External API-1	Google Gemini API used to generate FAQ content (questions, answers, topic suggestions).	Google Gemini API
9	External API-2	Not used in the current version.	N/A
10	Machine Learning Model	Generative language model used for automated FAQ and answer generation; hosted by Google and accessed through the API (no local training).	Google Gemini (LLM)
11	Infrastructure (Server / Cloud)	Application deployment on Local System / Cloud. Local Server Configuration: Node.js v18+, npm v9+, Windows 10/11 / macOS / Linux, Intel Core i5 (8th Gen) or AMD Ryzen 5, 8 GB RAM (16 GB recommended), 1 GB free storage. Cloud Server Configuration: any Node.js hosting platform with MongoDB Atlas (optional).	Local (VS Code, Node.js); Cloud hosting (optional)

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1	Open-Source Frameworks	Open-source frameworks and libraries used in the project.	Node.js, Express.js, Mongoose, cors, dotenv, jsonwebtoken, bcrypt
2	Security Implementations	Security and access controls: JWT-protected routes, password hashing, role-based authorization so only the author (or Admin) can modify an FAQ, CORS configuration, secrets kept in environment variables, and sanitized error responses that do not leak database details.	JWT, bcrypt, RBAC, CORS, dotenv, OWASP API Security guidelines
3	Scalable Architecture	Layered MVC architecture (routes, controllers, services, models) with a stateless JWT-based backend, so multiple server instances can run behind a load balancer as demand grows.	MVC, Express.js, MongoDB (replication / sharding)
4	Availability	Stateless server allows restart or redeploy without losing sessions; a managed, replicated database improves uptime when deployed on the cloud.	MongoDB Atlas replica set (optional), load balancer (optional)
5	Performance	Non-blocking asynchronous Node.js request handling, indexed MongoDB queries for fast FAQ search, and lightweight middleware.	Node.js async/await, MongoDB indexes

References:

1. C4 Model for software architecture: <https://c4model.com/>
2. IBM – Online order processing system during pandemic (pattern): <https://developer.ibm.com/patterns/online-order-processing-system-during-pandemic/>
3. IBM Cloud Architecture Center: <https://www.ibm.com/cloud/architecture>
4. AWS Architecture Center: <https://aws.amazon.com/architecture>
5. How to draw useful technical architecture diagrams: <https://medium.com/the-internal-startup/how-to-draw-useful-technical-architecture-diagrams-2d20c9fda90d>